

Loss functions in DL

Regression

MSE
MAE
number loss

Classification

binary cross entropy
categorical cross entropy
hinge loss

Autoencoders

KL divergence

GAN

discrimination loss
MinMax gan loss

object detection

focal loss

Embedding

Triplet loss

Loss function

↳ It is a method of evaluating how well your algorithm is modelling your dataset.

high → poor
low → good

It is find for every input

Loss function of linear regression

$$= y_i - \hat{y}_i$$

Actual	Pred
6.3	6.1
7.1	7.3
8.5	8.3

$$\text{Loss} = (6.3 - 6.1)^2$$

$$\text{Loss} = (7.1 - 7.3)^2$$

$$\text{Loss} = (8.5 - 8.3)^2$$

for each input

When whole training data sets loss is calculated it is known as cost function

$$\frac{1}{n} \sum (y_i - \hat{y}_i)^2$$

average of all

$$\frac{1}{3} [(6-1-6-3)^2 + (4-1-4)^2 + (3-5-3-7)^2]$$

= cost function

Loss function

1. Mean Squared Error (MSE) (Squared loss or L2 loss)

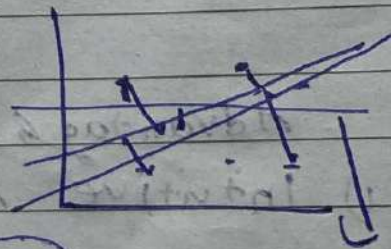
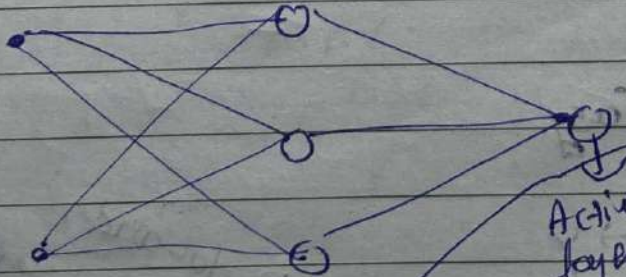
$$\frac{1}{n} \sum (y_i - \hat{y}_i)^2$$

True Predicted

$$\begin{aligned} 1 &= 1 \\ 2 &= 4 \\ 4 &= 16 \end{aligned}$$

Why this formula is not like $(y_i - \hat{y}_i)$ without square because the answer may be in -ve

Perceptron



Linear
 $y = wx + b$

$$CF = \frac{1}{n} \sum (y_i - \hat{y}_i)^2$$

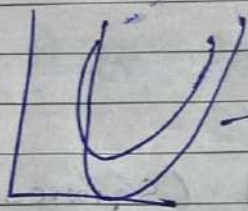
Activation function = linear
then only we can apply MSE
the more impact on line

Advantages

- 1) Easy to interpret
- 2) Differentiable (4D)
- 3) 1 local minima

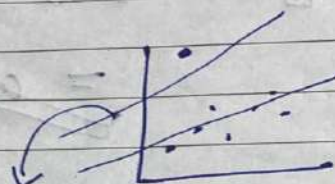
Disadvantages

- 1) Error Unit (squared) $\rightarrow$ diff unit of input
- 2) Robust to outliers



Graph
easy to find

have MSE fails to handle main data before of outliers
(Gradient Descent)

2. Mean Absolute Error (MAE) $\rightarrow$ L1 loss

$$L = |y_i - \hat{y}_i|$$

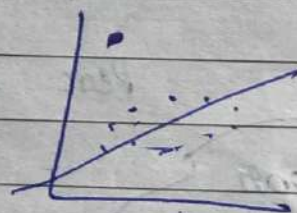
$$CF \Rightarrow \frac{1}{n} \sum |y_i - \hat{y}_i|$$

just change
Square with absolute

Advantages

- 1) Intuitive / Easy
- 2) Unit $\rightarrow$ same $\rightarrow y$
- 3) Robust to outliers

$\rightarrow$ No move towards outliers because
no square



MAE | MSE | Huber loss Regression outliers

PAGE NO.:

Dis Advantages

1) Not differentiable
↳ G.O.X

↳ calculate Subgradients

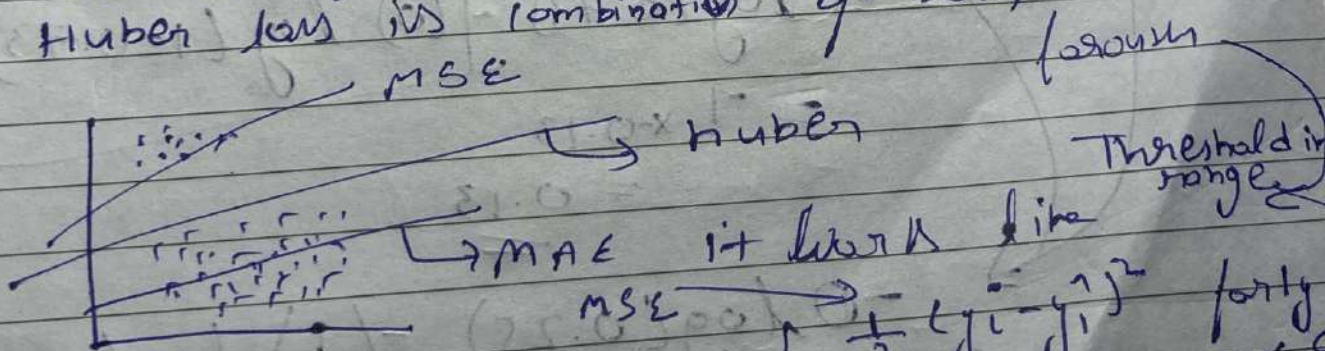
Note :- If data mein outliers nhi hai
chup chap use Krna MSE, nhi toh
MAE

3) Huber Huber loss

↓
Act like MSE &

If in data there are 25 outliers & 75 main points
if we use MSE it will heavy depends on

outliers & MAE will ignore outliers So
Huber loss is combination of both



MAE it looks like

$$L = \begin{cases} \frac{1}{2} (y_i - \hat{y}_i)^2 & \text{for } |y_i - \hat{y}_i| \leq \delta \\ |y_i - \hat{y}_i| - \frac{1}{2}\delta & \text{otherwise} \end{cases}$$

MAE

↳ outlier
range if exceeds
we MAE to
handle outliers

log loss

↳ Binary

↳ Classification

Activation

output = Sigmoid

$$\frac{1}{1+e^{-x}}$$

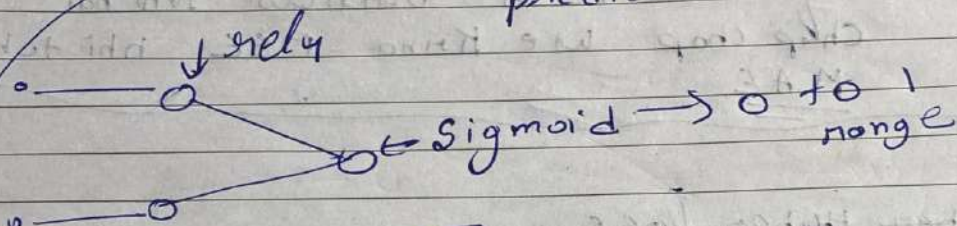
PAGE NO.:

When there is only 2 outputs (1/0) (cat, dog)

Actual value

$$LF = -y \log(y^{\hat{}}) - (1-y) \log(1-y^{\hat{}})$$

prediction



$$CF = -\frac{1}{n} \left[\sum_{i=1}^n y_i \log(y_i^{\hat{}}) + (1-y_i) \log(1-y_i^{\hat{}}) \right]$$

gpx	i4	prob
7	80	1
70	70	0

$$-1 \log(0.73) - (0) \log(1-0.73)$$

$$= -1 \times 0.13$$
$$= 0.13$$

$$-0 \log(0.25) - (1) \log(0.75)$$

$$= -1 \times 0.12$$
$$= 0.12$$

Advantage

Disadvantage

* Differentiable

multiple local minima
Intuitive & not easy

5. Categorical Cross Entropy

Output activation = Softmax

Kan classes

We use one hot encoding
no of classes

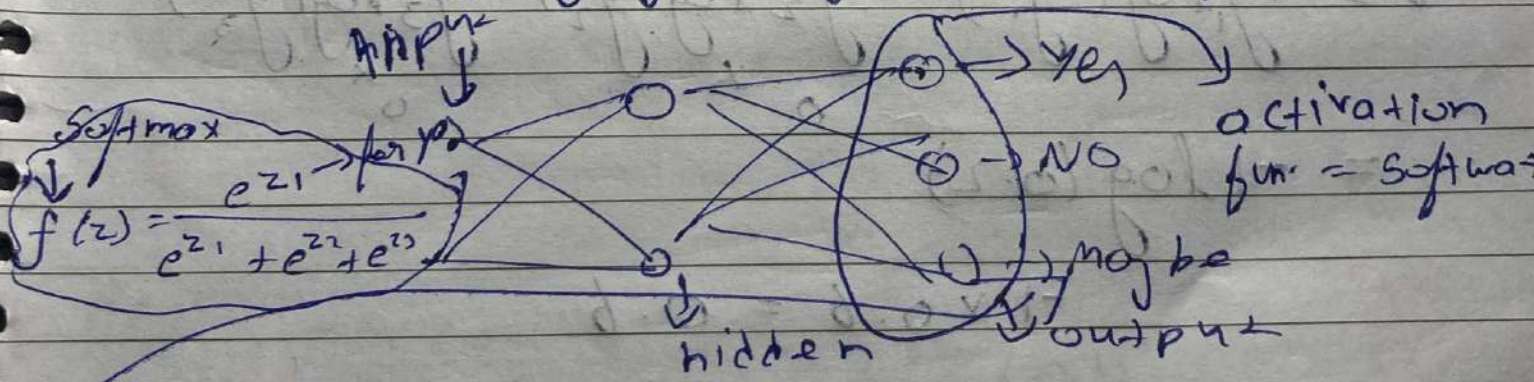
multiclass

outputs
Yes
NO
maybe

$$L = - \sum_{j=1}^K y_i \log(\hat{y}_j)$$

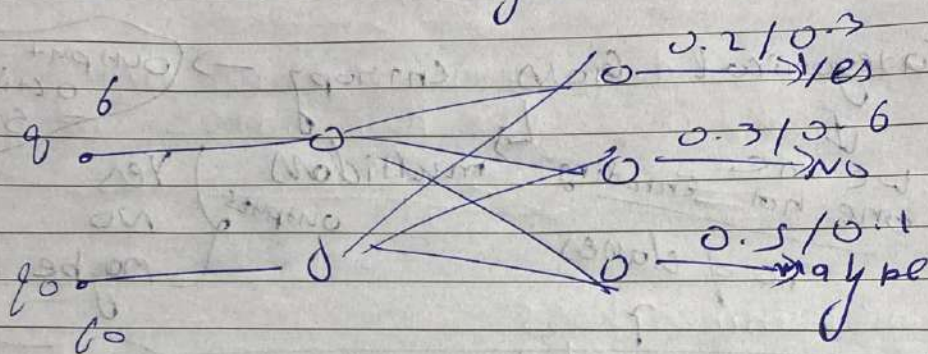
$$CF = -\frac{1}{n} \sum_{i=1}^n \sum_{j=1}^K y_{ij} \log(\hat{y}_{ij})$$

$$L = -y_1 \log(\hat{y}_1) - y_2 \log(\hat{y}_2) - y_3 \log(\hat{y}_3)$$



Jbhi ap Ek multiclass pe kaam kr rhe ho
apka neural network ki structure ko
uthi hae classes ko neurons bnata h.
for eg → Yes, no, maybe

Ques	is placed	Yes	No	my
1	for	Yes	1	0
2	for	No	0	1
3	for	my	0	0



$$[0.2, 0.3, 0.5]$$

$$[1, 0, 0]$$

$$L = -y_1 \log(y_1) - y_2 \log(y_2) - y_3 \log(y_3)$$

$$= -1 \log(0.2)$$

$$-1 \times 0.6 = 0.6$$

$$L = [0.3, 0.6, 0.1]$$

$$[0, 1, 1]$$

$$L = -y_2 \log(y_2)$$

$$= -1 \times \log(0.6)$$

$$= -1 \times 0.21 = 0.21$$

6. Sparse ^{from} categorical Entropy

↓
Some formula
but only
calculated for
actual output

Every thing same as categorical
Cross Entropy but

but we don't use one hot encoding
we use integer encoding

cgpa	id	place	OHc
7	70	Yes	1
8	80	No	2
6	60	Maybe	3

↓
[0.1, 0.4, 0.5] $\frac{1}{\text{actual}}$

$$L = -1 \log(0.5)$$

[0.1, 0.4, 0.6] $\frac{2}{\text{actual}}$

Reg $\begin{cases} \rightarrow \text{MSE} \\ \rightarrow \text{MAE (outlier)} \\ \rightarrow \text{huber} \end{cases}$
 $L = -1 \log(0.4)$
 (many outliers)

Classification $\begin{cases} \rightarrow \text{Binary cross entropy} \rightarrow \text{2 class} \\ \rightarrow \text{categorical cross entropy} \rightarrow \text{multi class} \\ \rightarrow \text{Sparse categorical cross entropy} \rightarrow \text{multi class} \end{cases}$